

Dear Members of the Search Committee,

I am pleased to write this letter in strong support of **Dr. Jialun Cao**, whom I had the opportunity to evaluate as a **member of her PhD dissertation examination committee**. In this role, I conducted an independent and critical assessment of her doctoral dissertation, titled “Towards Automatic Testing and Fault Localization in Natural Language Processing Systems” and research contributions, including a careful review of the thesis manuscript and her oral defense. Based on this evaluation, I can state unambiguously that Dr. Cao’s doctoral work is **outstanding by international standards**.

Dr. Cao’s dissertation makes **exceptional contributions to software engineering**, particularly at the intersection of **artificial intelligence, software testing, and program verification**. What distinguishes her work is not only its technical correctness, but also its **conceptual clarity, methodological rigor, and sustained research depth across multiple interrelated problems**. In my experience as a PhD examiner, dissertations of this caliber are rare.

The originality and impact of her doctoral research were formally recognized by the **ACM SIGSOFT 2025 Outstanding Dissertation Award**, the most prestigious doctoral-level award in the global software engineering community and one that is typically conferred on **only a single recipient worldwide each year**. Having examined the substance of her dissertation in detail, I can confidently say that this recognition is fully deserved.

One of the central contributions of her dissertation is the **DeepFD framework** (ICSE 2022; JASE 2024), which advances the state of the art in automated fault localization and diagnosis for deep learning programs. From an examiner’s perspective, this work stands out for its clear problem formulation, strong technical innovation, and rigorous empirical evaluation. Most notably, it **reconceptualizes fault localization in deep learning** by moving beyond the prevailing notion of identifying faulty neurons, and instead **formulates the problem at the level of deep learning programs themselves**. This shift establishes a **new paradigm** for understanding, diagnosing, and debugging failures in deep neural systems.

Beyond DeepFD, Dr. Cao’s dissertation presents a **coherent and forward-looking research agenda** rather than a collection of loosely connected papers. In particular, her work on **program specification synthesis using large language models** (CAV 2024; ACL Main Conference 2025) represents a significant conceptual advance in bridging informal requirements and formal verification. These contributions demonstrate a sophisticated understanding of both **formal methods and modern AI techniques**, and they push the boundary of what is currently possible in program analysis and verification.

From the perspective of a dissertation examiner, I was particularly impressed by Dr. Cao’s ability to **identify fundamental research questions**, position her work clearly within the broader literature, and articulate its long-term significance to the field. The level of intellectual independence and research maturity reflected in her dissertation is **well above what I would typically expect from a PhD candidate**, and is more comparable to that of a strong early-career faculty member.

Dr. Cao's research record further reinforces this assessment. She has published extensively in the **most selective venues** in software engineering and related areas, including **ICSE, FSE, ASE, TOSEM, and USENIX Security**, and her work has already accumulated **over 700 citations**. Taken together with the depth and coherence of her dissertation, this record places her among the **most outstanding recent PhD graduates** in software engineering internationally.

In summary, based on my role as a **PhD dissertation examiner**, I consider Dr. Jialun Cao's doctoral research to be **exceptional in originality, technical depth, and impact**. Her dissertation meets, and in many respects exceeds, the standard expected of award-winning doctoral work at the international level. I have no hesitation in stating that she possesses the research excellence and intellectual independence necessary to succeed as an **Assistant Professor**.

I recommend Dr. Jialun Cao with the highest confidence and without reservation. Please feel free to contact me if further information would be helpful.

Sincerely,

A handwritten signature in black ink, appearing to read 'Mariani Leonardo', with a stylized, cursive script.

Mariani Leonardo
Full Professor and Director of the Ph.D. Program
University of Milano - Bicocca